Diode models

Answers and worked solutions · Day 03 · Mon 26 Oct · pack E2

5. Practice

From your course files

WS2 Q8 (course question).

> The official solution uses $V_T = 25$ mV (its "V_T at 20 °C ≈ 25 mV") and $n = 1$. At mA currents $I_D \gg I_S$, so $V_D = nV_T \ln(I_D/I_S)$. (The full equation with the "−1" gives the same 0.5583 V at 1 mA, which confirms the diode is fully conducting.)
>
> - **i.** $V_D = 0.025 \ln \dfrac{10^{-3}}{0.2 \times 10^{-12}} = 0.025 \times 22.33 =$ **0.5583 V**
> - **ii.** $V_D = 0.025 \ln(5 \times 10^{10}) =$ **0.6159 V**
> - **iii.** $V_D = 0.025 \ln(5 \times 10^{11}) =$ **0.6734 V**
> - **iv.** Each decade adds $\Delta V = nV_T \ln 10 = 2.303 \times 25\ \text{mV} =$ **57.6 mV**. With $V_T = 25.9$ mV (27 °C) it is about 60 mV per decade.

WS2 Q9 (course question).

> Worked in full in Example 1 above (redo it closed-book). Dividing the two operating points removes I_S: $0.1 = e^{-0.1/(n \times 0.025)}$, so $n = 1.737$, and then $I_S = 10^{-3} e^{-0.7/(1.737 \times 0.025)} =$ **0.1 nA**.

WS2 Q10 (course question).

> Identical diodes share I equally, so each now carries $I/2$. From $I = I_S e^{V_1/nV_T}$ and $I/2 = I_S e^{V_2/nV_T}$:
>
> $$\frac{1}{2} = e^{(V_2 - V_1)/nV_T} \;\Rightarrow\; V_2 - V_1 = -nV_T \ln 2$$
>
> The voltage **falls** by $nV_T \ln 2$, about 17–18 mV for $n = 1$.

WS2 Q11 (course question).

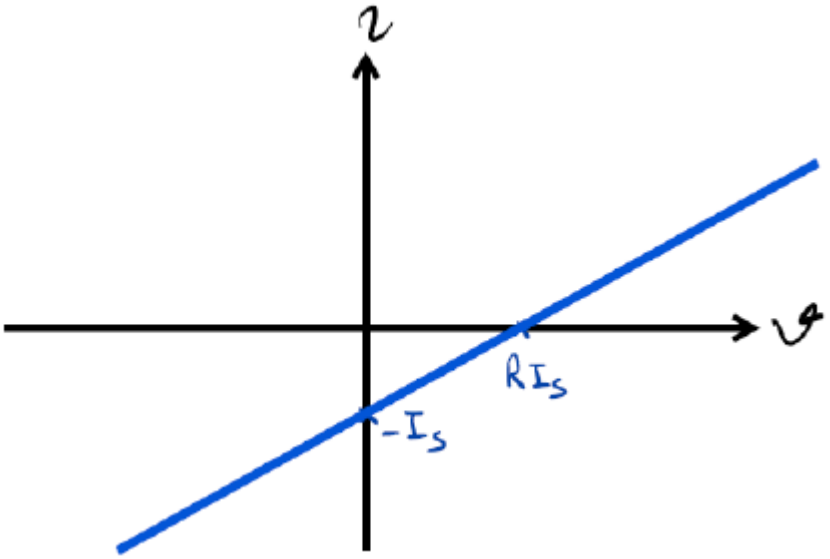
> [A] **anode**, [B] **p-type**, [C] **boron** (or another trivalent, Group III atom), [D] **holes**, [E] **electrons**.

WS2 Q12 (course question).

Yes, both are linear. Each contains only linear elements (a resistor and an independent source), so by superposition the terminal i–v relation is a straight line. Here I_S is an independent current source, not a diode's saturation current.

WS2 Q13 (course question).

Circuit A: KCL at the top node, $i + I_S = I_R = v/R$, so $i = \dfrac{v}{R} - I_S$. Intercepts: $i = 0$ at $v = RI_S$; $v = 0$ at $i = -I_S$.



WS2 Q13 official sketch for circuit A

Circuit B: KCL, $I_S = i + v/R$, so $i = I_S - \dfrac{v}{R}$. Intercepts: $i = 0$ at $v = RI_S$; $v = 0$ at $i = +I_S$. A straight line of slope $-1/R$.

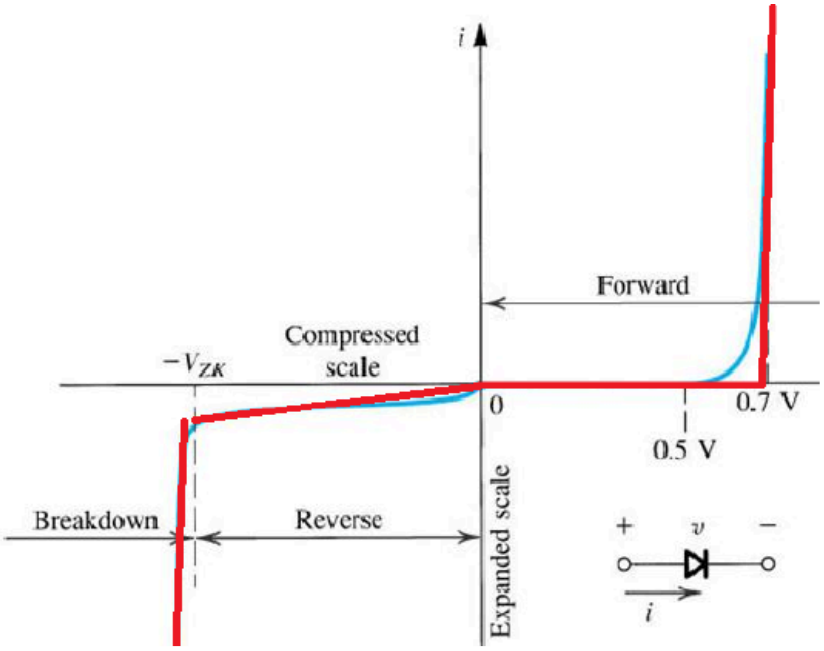
Both are straight lines, confirming Q12.

WS2 Q14 (course question).

- Region A: **breakdown**. Region B: **reverse bias** (only I_S flows). Region C: **partial conduction**. Region D: **full conduction**.
- Voltage A: **Zener knee / breakdown voltage**. Voltage B: **cut-in voltage**. Voltage C: **forward voltage drop**.

WS2 Q15 (course question).

Four linear pieces, one per region (breakdown, reverse, partial conduction, full conduction), because each region behaves differently and no single straight line fits two of them.



WS2 Q15 official sketch: four straight pieces (red) over the real curve

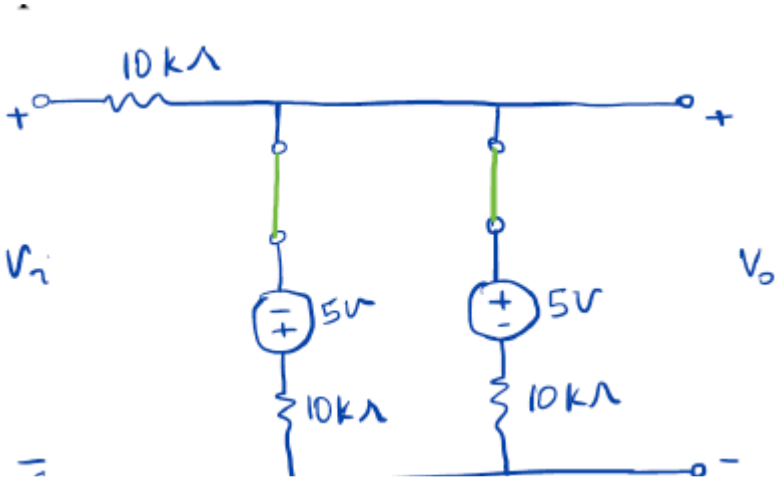
WS2 Q16 (course question).

The ohmmeter's own source is **reverse biasing** the diode, so it reads the very large reverse resistance (almost an open circuit). **Nothing is wrong.** Rotate the diode and it is forward biased, so you expect a **very low** reading (a few to a few tens of ohms). This is a quick way to find a diode's polarity, but the forward reading is at one unknown operating point, so it is not the diode's forward resistance r_D.

WS2 Q17 (course question).

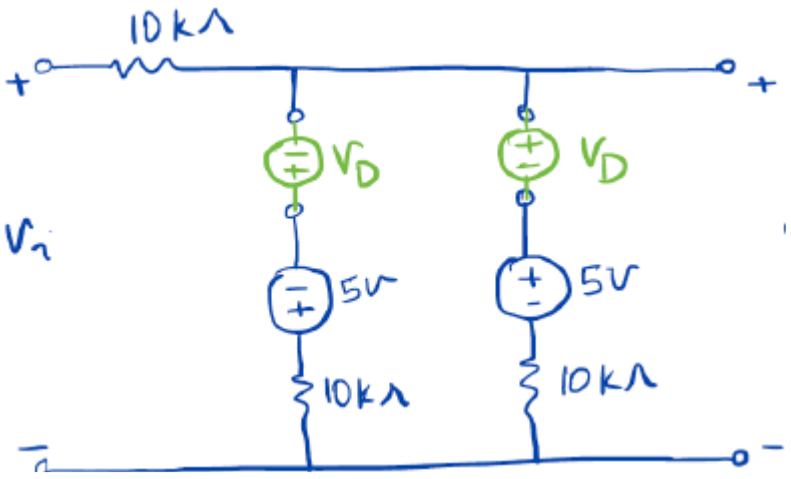
Redraw only. Both diodes are assumed ON, so each is replaced in turn by the forward model.

1. Ideal: each diode becomes a short circuit (green).



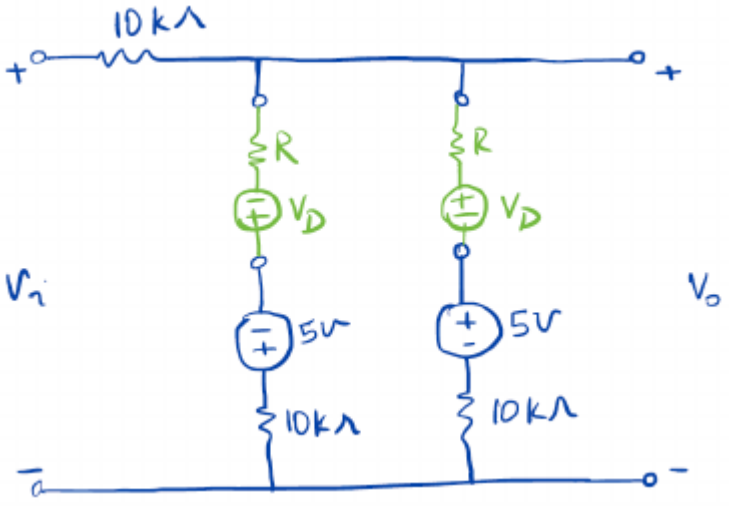
WS2 Q17 official redraw, ideal model

2. Voltage drop: each diode becomes a V_D source with + on the anode side.



WS2 Q17 official redraw, voltage-drop model

3. $V_D + R$**:** each diode becomes V_D in series with R (that is, r_D).



WS2 Q17 official redraw, voltage drop with resistor

In reverse bias every piecewise-linear model is just an open circuit.

WS3 Q1 (course question).

$T = 298.15$ K, so $V_T = \dfrac{1.38 \times 10^{-23} \times 298.15}{1.6 \times 10^{-19}} = 25.72$ mV. With $I_D \gg I_S$:

$$V = nV_T \ln \frac{I_D}{I_S} = 25.72\ \text{mV} \times \ln \frac{100 \times 10^{-3}}{10^{-6}} = 25.72 \times 11.51 = \mathbf{296\ mV}$$

(official 296.11 mV; see the note above).

WS3 Q2 (course question).

The diode is reverse biased, so the current round the loop is I_S and $V = I_S \times 1\ \text{M}\Omega$. At 20 °C: $I_S = 1/10^6 = 1\ \mu$A.

- 40 °C is +20 °C, two doublings: $I_S = 4\ \mu$A, so $V = 4$ **V**.
- 0 °C is −20 °C, two halvings: $I_S = 0.25\ \mu$A, so $V = 0.25$ **V**.

WS3 Q3 (course question).

Each diode must drop $2.4/3 = 0.8$ V, which is 0.1 V above 0.7 V, so one decade above 1 mA: $I_D = 10$ mA.

$$R = \frac{10 - 2.4}{10\ \text{mA}} = \mathbf{760\ \Omega}$$

WS3 Q4 (course question).

$T = 293$ K: $V_T = \dfrac{1.38 \times 10^{-23} \times 293}{1.6 \times 10^{-19}} =$ **25.27 mV**. Full equation:

$$I_D = I_S\left(e^{V_D/nV_T} - 1\right) = 40 \times 10^{-9}\left(e^{0.5/(2\times 0.02527)} - 1\right) = \mathbf{0.79\ mA}$$

(official 0.789 mA; I get 0.791 mA).

WS3 Q5 (course question).

Use the knee voltage for the source and the chord between two points on the steep part for $r_{av} = \Delta V_d / \Delta I_d$ (the official solution uses 4 mA and 30 mA).

- **Ge:** $V_K = 0.3$ V, $r_{av} = \dfrac{0.42 - 0.3}{(30 - 4)\ \text{mA}} = 4.62\ \Omega$. Model: ideal diode + **0.3 V** + **4.62 Ω**.
- **GaAs:** $V_K = 1.2$ V, $r_{av} = \dfrac{1.32 - 1.2}{(30 - 4)\ \text{mA}} = 4.62\ \Omega$. Model: ideal diode + **1.2 V** + **4.62 Ω**.

WS3 Q6 (course question).

(a) At 4 mA the graph gives $V_D \approx 0.7$ V, so $R_D = V_D / I_D = 0.7/4\ \text{mA} =$ **175 Ω**.

(b) Draw the tangent at 10 mA and read ΔV_D and ΔI_D off it. The official answer is $r_d \approx 4$ **Ω** (see the note at the top of this section about its printed ΔI_D).

WS3 Q7 (course question).

(a) Both diodes conduct (Si 0.7 V, Ge 0.3 V). Divider on what is left:

$$V_o = \frac{10}{10+2}(12 - 0.7 - 0.3) = \mathbf{9.167\ V}$$

(b) The top Si diode and the parallel pair (which drops only one 0.7 V) are on. Current through 4.7 kΩ:

$$I = \frac{16 - 0.7 - 0.7 - (-4)}{4.7\ \text{k}} = 3.96\ \text{mA}, \qquad V_o = -4 + 4.7\ \text{k} \times 3.96\ \text{mA} = \mathbf{14.6\ V}$$

WS3 Q8 (course question).

Assume the diode is **OFF**. Then both sides are plain dividers:

$$V_A = 5 \cdot \frac{2}{2+3} = \mathbf{2\ V}, \qquad V_B = 5 \cdot \frac{2}{2+2} = \mathbf{2.5\ V}$$

Check: $v_d = V_A - V_B = -0.5$ V, which is below the 0.7 V threshold (it is reverse biased), so **OFF** is correct.

PT Q2 (practice test, 5 marks).

Step 1. Thermal voltage. $T_K = 273 + 27 = 300$ K.

$$V_T = \frac{kT_K}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875\ \text{mV}$$

Step 2. Ratio form. Both currents are much larger than I_S, so $I_D \simeq I_S e^{V_D/nV_T}$. Divide the two operating points:

$$\frac{I_2}{I_1} = e^{(V_2 - V_1)/nV_T} \;\Rightarrow\; \Delta V_D = nV_T \ln \frac{I_2}{I_1}$$

I_S cancels, which is why the question does not give it.

Step 3. Substitute.

$$\Delta V_D = 1.5 \times 25.875 \times 10^{-3} \times \ln \frac{10}{0.1} = 1.5 \times 0.025875 \times 4.605 = \boxed{0.179\ \text{V}\ (179\ \text{mV})}$$

The voltage **increases** by 179 mV.

> **SANITY CHECK:** one decade costs $2.303 \times 1.5 \times 25.875 = 89.4$ mV, and two decades cost 179 mV ✓.

PT Q3 (practice test, 8 marks).

READING THE FIGURE. The 10 V source feeds a 5 kΩ series resistor to node A. A 5 kΩ resistor runs from A to ground. The three diodes run from A to node B, all pointing from A to B. From B, 10 kΩ goes down to the + terminal of a 3 V source whose − terminal is grounded. V_o is measured from B to ground.

(a) Redraw. Replace each diode with an ideal diode in series with a 0.6 V source (+ on the anode side, so it opposes forward current) and a 0.2 kΩ resistor. Draw all three separately, because the question says not to simplify:

```
10 V o--[5k]--+--|>|-(+0.6V-)-[0.2k]--|>|-(+0.6V-)-[0.2k]--|>|-(+0.6V-)-[0.2k]--+--o
              A          D1                        D2                       D3      B
              |                                                                    |
            [5k]                                                                [10k]
              |                                                                    |
             GND                                                          (+) 3 V (-)
                                                                                   |
                                                                                  GND
```

Practice test Q3(a) redrawn: each diode replaced by an ideal diode, a 0.6 V source (+ on the anode side) and 0.2 kΩ.

Label the chain current I, flowing from A to B (anode to cathode).

(b) Verify. Work in V, mA and kΩ.

1. Model: 3rd model, all three ON (as the question tells you to assume).
2. KCL at A: $\dfrac{10 - V_A}{5} = \dfrac{V_A}{5} + I \;\Rightarrow\; V_A = 5 - 2.5I$.
3. Branch B to ground: $V_B = 3 + 10I$.
4. Diode chain: $V_A - V_B = 3(0.6) + 3(0.2)I = 1.8 + 0.6I$.
5. Substitute: $(5 - 2.5I) - (3 + 10I) = 1.8 + 0.6I \;\Rightarrow\; 0.2 = 13.1I$

$$I = 0.01527\ \text{mA} = 15.3\ \mu\text{A}$$

The cross-check by Thévenin is quicker. The source, the 5 k and the 5 k seen from A give $V_{Th} = 5$ V and $R_{Th} = 2.5$ kΩ. Then $I = \dfrac{5 - 1.8 - 3}{2.5 + 0.6 + 10} = \dfrac{0.2}{13.1} = 15.3$ μA ✓.

CHECK: $I = 15.3$ μA > 0 in the anode-to-cathode direction. Because the diodes are in series, the same positive current flows through each one, so **all three diodes are conducting** ✓. The node voltages are $V_A = 4.962$ V and $V_B = 3.153$ V. The chain drops 1.809 V, which equals $1.8 + 0.6(0.0153)$ ✓.

(c) Output voltage.

$$V_o = V_B = 3 + 10\ \text{k}\Omega \times 15.3\ \mu\text{A} = \boxed{3.15\ \text{V}}$$

Comment worth writing: the margin is tiny. Only 0.2 V is left after the three 0.6 V drops and the 3 V source. With the 2nd model at 0.7 V per diode, $5 - 2.1 - 3 = -0.1$ V would mean the diodes are OFF and $V_o = 3$ V. That is why the verification part is worth 4 marks.

New questions

N1 [4 marks]

a. $T_K = 323$ K, so $V_T = 1.38 \times 10^{-23} \times 323 / 1.6 \times 10^{-19} = \mathbf{27.86\ mV}$.

b. $\Delta V_D = nV_T \ln(I_2/I_1) = 1.8 \times 0.02786 \times \ln(0.1) = -0.1155$ V. So $V_D = 0.62 - 0.1155 = \mathbf{0.505\ V}$. One decade costs $2.303 nV_T = 115$ mV ✓.

c. $r_d = nV_T / I_D = 1.8 \times 27.86\ \text{mV} / 2\ \text{mA} = \mathbf{25.1\ \Omega}$.

N2 [6 marks]

a. $V_T = 25.875$ mV. Dividing the two operating points gives

$$n = \frac{0.69 - 0.60}{0.025875 \ln 10} = \mathbf{1.511}$$

Then $I_S = 0.5 \times 10^{-3} e^{-0.60/(1.511 \times 0.025875)} = \mathbf{1.08 \times 10^{-10}\ A}$ (0.108 nA). The second point gives the same value ✓.

b. $V_D = 0.69 + nV_T \ln(20/5) = 0.69 + 1.511 \times 0.025875 \times 1.386 = \mathbf{0.744\ V}$.

c. $R_D = 0.69/5\ \text{mA} = \mathbf{138\ \Omega}$. $r_d = 1.511 \times 25.875\ \text{mV}/5\ \text{mA} = \mathbf{7.82\ \Omega}$. A 2 mV signal is below about 5 mV, so it sees r_d.

N3 [4 marks]

a. $V_D \approx 0.65 - 2.5\ \text{mV} \times 50 = \mathbf{0.525\ V}$.

b. At 75 °C: $10\ \text{nA} \times 2^{50/10} = 10 \times 32 = \mathbf{320\ nA}$. At 0 °C: $10 \times 2^{-25/10} = \mathbf{1.77\ nA}$.

c. $V_T(25\ ^\circ\text{C}) = 25.70$ mV, so $r_d = 25.7\ \Omega$. $V_T(75\ ^\circ\text{C}) = 30.02$ mV, so $r_d = 30.0\ \Omega$. (V_T here uses $T_K = T + 273$.)

N4 [7 marks] (multi-diode, iterate)

a. **Assume D1 ON and D2 ON.** Write KCL at A, in mA with kΩ:

$$\frac{10 - V_A}{1} = \frac{V_A - 0.7}{2} + \frac{V_A - 0.7 - 8}{1} \;\Rightarrow\; 19.05 = 2.5V_A \;\Rightarrow\; V_A = 7.62 \text{ V}$$

This gives $i_{d1} = (7.62 - 0.7)/2 = 3.46$ mA > 0 ✓, but $i_{d2} = (7.62 - 8.7)/1 = -1.08$ mA < 0 ×. **D2 cannot be ON.** The 8 V source pushes back harder than node A can push forward.

b. **Assume D1 ON and D2 OFF.** D2 is now an open circuit.

$$I = \frac{10 - 0.7}{1 + 2} = 3.10 \text{ mA}, \quad V_A = 10 - 3.10 = 6.90 \text{ V}, \quad V_B = 6.90 - 0.7 = 6.20 \text{ V}$$

No current flows in the 1 kΩ of D2's branch, so D2's cathode sits at 8 V. That gives $v_{d2} = 6.90 - 8 = -1.10$ V.

Check: $i_{d1} = 3.10$ mA > 0 ✓ and $v_{d2} = -1.10$ V < 0.7 V ✓.

$$\boxed{V_o = 6.20 \text{ V}} \quad (\text{D1 ON, D2 OFF})$$

N5 [8 marks] (two diodes, two input polarities, chained)

a. **Assume D1 ON and D2 OFF.** D1's anode faces $+6$ V. D2's cathode is at the positive node. Branch 1 is 0.7 V + 2 V in series with $0.1 + 0.5 = 0.6$ kΩ, so its current is $(V_A - 2.7)/0.6$. KCL at A:

$$\frac{6 - V_A}{1} = \frac{V_A}{4} + \frac{V_A - 2.7}{0.6} \;\Rightarrow\; V_A = 3.60 \text{ V}$$

That gives $i_{d1} = (3.60 - 2.7)/0.6 = 1.50$ mA > 0 ✓. D2's anode is at 0 V (no current in its 1 kΩ) and its cathode is at 3.60 V, so $v_{d2} = -3.60$ V < 0.7 ✓. As a KCL check, $2.40 = 0.90 + 1.50$ mA ✓. $V_o = 3.60$ **V.**

b. **Assume D2 ON and D1 OFF.** Current flows up from ground through 1 kΩ and D2 into A, then through R to the -6 V source. The current into A through D2 is $(0 - 0.7 - V_A)/1.1$. KCL at A (current in through R = current out through the 4 k + current out through branch 2, which is negative here):

$$\frac{-6 - V_A}{1} = \frac{V_A}{4} - \frac{-0.7 - V_A}{1.1} \;\Rightarrow\; V_A = -3.07 \text{ V}$$

That gives $i_{d2} = (-0.7 + 3.074)/1.1 = 2.16$ mA > 0 ✓. D1's cathode is held at +2 V by the battery (no current in 0.5 kΩ), so $v_{d1} = -3.07 - 2 = -5.07$ V < 0.7 ✓. $V_o = -3.07$ **V.**

c. With both diodes OFF, $V_A = V_i \times \dfrac{4}{1 + 4} = 0.8V_i$.

- D1 turns on when $v_{d1} = V_A - 2 = 0.7$, that is $V_A = 2.7$ V, so $V_i = 3.375$ V.
- D2 turns on when $v_{d2} = 0 - V_A = 0.7$, that is $V_A = -0.7$ V, so $V_i = -0.875$ V.

Both diodes are OFF for $-0.875\ \text{V} < V_i < 3.375\ \text{V}$. Parts (a) and (b) lie outside this range, which is consistent.

N6 [8 marks] (OR gate with unequal inputs: the near-miss)

Each ON branch is a source $V_k - 0.7$ in series with $1 + 0.1 = 1.1$ kΩ.

a. **Assume both ON.** KCL at V_o:

$$\frac{4.3 - V_o}{1.1} + \frac{3.3 - V_o}{1.1} = \frac{V_o}{4} \;\Rightarrow\; V_o = 3.341\ \text{V}$$

That gives $i_{d1} = 0.872$ mA > 0 ✓, but $i_{d2} = (3.3 - 3.341)/1.1 = -0.037$ mA < 0 ×. **D2 is not ON.** The failure is small, so it is easy to miss if you round early.

b. **Assume D1 ON and D2 OFF.**

$$i_{d1} = \frac{4.3}{1.1 + 4} = 0.843\ \text{mA}, \qquad V_o = 4 \times 0.843 = 3.373\ \text{V}$$

No current flows in D2's 1 kΩ, so its anode is at 4 V and $v_{d2} = 4 - 3.373 = 0.627$ V.

Check: $i_{d1} > 0$ ✓ and $v_{d2} = 0.627$ V < 0.7 V ✓. $V_o = 3.37$ **V.** D2 is forward biased, but by less than V_D, so in this model it is OFF.

c. D2 starts to conduct when $v_{d2} = 0.7$ V with zero current, that is $V_2 = 3.373 + 0.7 = \mathbf{4.07\ V}$.

d. $V_2 = 5$ V is symmetric with V_1, so both are ON with equal currents: $2(4.3 - V_o)/1.1 = V_o/4$. That gives $V_o = \mathbf{3.78\ V}$, with $i_{d1} = i_{d2} = 0.473$ mA > 0 ✓.

6. Quick recall check

Question

1. $V_T \ln 10 = 57.6$ mV. With $n = 2$ it is 115 mV.
2. It falls, by $nV_T \ln 2 \approx 18$ mV for $n = 1$.
3. An open circuit in all three models.
4. $26/13 = 2\ \Omega$.
5. $5 \times 2^2 = 20$ nA. The forward drop falls by about $2.5 \times 20 = 50$ mV at a fixed current.